Universität Trier, Fachbereich II
Professur Computerlinguistik

Layer Wise Gender Bias in Multilingual Foundation Models: A Five Architecture Cross Lingual Analysis

Machine Learning for Natural Language Understanding (MA2NLP2001)
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submitted by

Juan Jose Arango Serrano
Matr. 1861694
s2juaran@uni-trier.de

Hadil Athmouni
Matr. 1762515
s2haathm@uni-trier.de

Daniel Tesfai Kebede
Matr. 1716694
s4dakebe@uni-trier.de

Instructor: Raghvi Baloni

Abstract

Multilingual foundation models represent a central trend in modern machine learning, yet pretraining across massive corpora often encodes social biases. While previous studies have demonstrated that static embeddings distilled from contextualized models encode gender bias in English, Spanish presents a particular challenge as most professions are morphologically gender marked, confusing grammatical gender with social bias.

This study evaluates how static embeddings extracted from multilingual models assign gender bias to grammatically invariant Spanish professions (e.g., *dentista*, *taxista*) without explicit supervision. Comparative analysis is conducted across Arabic and Tigrina.

Using geometric test across multiple architectures, we demonstrate that while models successfully encode the grammatical neutrality of certain suffixes, statistical frequency and societal bias override morphological rules for fields historically dominated by men. Finally, our results demonstrate that the internal design of the model also has an influence bias direction.

1 Introduction

Large language models trained across numerous languages represent a central trend in modern machine learning. Models such as mBERT and the XLM RoBERTa family create shared representational spaces for dozens of languages. This compression raises an important structural question, when languages with radically unequal training data coexist, how do these models distribute social associations across them? Bommasani et al. (2020) demonstrated that static embeddings distilled from English contextualized models encode social biases. However, English nouns generally lack grammatical gender. In Spanish, is very common to find nouns marked by gender (e.g., *maestro/maestra*), which often confuses grammatical gender with social bias.

This research investigates how these models handle grammatically neutral Spanish professions compared to those with explicit gender markers. We also place Spanish within a broader comparative context alongside languages with lower training resources, such as Tigrinya and Arabic. By examining these models across their entire depth, we aim to understand how gender bias varies from the initial input to the final output. Previous studies establish that early layers process syntax while middle layers encode semantic relationships, which is exactly where we expect social associations to peak before a final representational collapse at the output layer.

Furthermore, we evaluate the robustness of our geometric bias measurements. We test whether a projection estimator sensitive to magnitude and a cosine estimator based entirely on angles agree on the underlying trends. We observed a Spearman correlation exceeding 0.96 across all valid models, confirming the reliability of our mathematical approach.

This research aims to fill the identified gap with a layer wise, cross linguistic and architecture comparative, with Spanish as the main focus, following the questions:

- **RQ1**: Do multilingual transformers encode measurable gender bias in Tigrigna, Arabic, and Spanish under a unified geometric protocol?
- **RQ2**: How does measured bias vary across transformer layers, and at which depths do associations

peak?

- **RQ3**: Do two independent geometric estimators agree on bias trends across models?
- **RQ4**: Does attention architecture determine gender bias direction, independent of pretraining data?

2 Related Work

2.1 Bias in Static Word Embeddings

Early work established that social biases are encoded geometrically in static word embeddings. Bolukbasi et al. (2016) derived a gender direction from anchor sets, revealing systematic profession gender associations and proposing projection based debiasing. Caliskan et al. (2017) introduced WEAT as a statistical framework for implicit associations; Garg et al. (2018) showed embedding bias correlates with historical trends. Gonen and Goldberg (2019) demonstrated that removing a single bias direction does not eliminate deeper geometric bias, motivating continued geometric analysis.

2.2 Interpreting Contextual Representations via Reductions

Transformer models produce contextualised representations that vary across inputs and depth, complicating direct application of geometric methods. Bommasani et al. (2020) formalised reductions of contextual embeddings to stable word level vectors suitable for geometric analysis, while Rogers et al. (2020) surveys the unequal distribution of linguistic information across transformer layers, motivating layer wise analysis.

2.3 Multilingual Bias and Cross Lingual Considerations

As multilingual transformers became widely adopted, bias analysis extended beyond English. Zhao et al. (2019) and Zhao et al. (2020) showed that geometric bias methods transfer to contextual multilingual models and that bias patterns vary substantially across languages, with debiasing in one language not generalising to others. Guo and Caliskan (2021) further showed contextualised embeddings encode intersectional associations. These studies predominantly evaluate final layer representations. Lauscher et al. (2020) showed zero shot cross lingual transfer degrades substantially for typologically distant and low resource languages.

2.4 Architecture Driven Effects on Representations

The models evaluated here differ not only in scale but in fundamental attention design. He et al. (2021) introduced Disentangled Attention, which separates content and positional information into independent attention matrices, substantially changing how contextual relationships are encoded across layers. He et al. (2023) extended this design to multilingual pretraining via an ELECTRA-style training objective, producing the microsoft/mdeberta-v3-base checkpoint evaluated here. Liang et al. (2023) enlarged the Sentence Piece vocabulary from 250K tokens (XLM-R) to 1M tokens (XLM-V) to improve tokenisation

coverage for low resource languages. These architectural and vocabulary decisions constitute the design space whose bias consequences this study directly compares.

2.5 How This Work Differs

We integrate geometric bias analysis Bolukbasi et al. (2016) with contextual-to-static reductions Bommasani et al. (2020) in a controlled layer wise, architecture comparative study across five multilingual models, revealing that Disentangled Attention produces a complete reversal of gender bias direction not previously documented in the multilingual bias literature.

3 Methodology

3.1 Models

We evaluate four pretrained multilingual transformer models without fine tuning. All checkpoints are loaded from HuggingFace Hub; no parameter updates are performed.

- **mBERT** (`bert-base-multilingual-cased`; Devlin et al., 2019): A 12-layer Transformer model (yielding 13 hidden states including the embedding layer). It employs a WordPiece vocabulary of approximately 120K tokens covering 104 languages. It serves as our baseline for multilingual pre-training.
- **XLM-RoBERTa-base** (`xlm-roberta-base`; Conneau et al., 2020): Consists of 12 layers (13 hidden states) with a 250K-token SentencePiece vocabulary. It was trained on 2.5TB of CommonCrawl data, significantly scaling the training data compared to mBERT.
- **XLM-RoBERTa-large** (`xlm-roberta-large`; Conneau et al., 2020): A larger variant featuring 24 layers (25 hidden states) and a hidden dimension of 1024. It utilizes the same 250K SentencePiece vocabulary as the base model but provides higher capacity for complex linguistic patterns.
- **XLM-V-base** (`facebook/xlm-v-base`; Liang et al., 2023): This model maintains a 12-layer architecture (13 hidden states) but features an expanded SentencePiece vocabulary of 1M tokens. This design choice aims to solve the "vocabulary bottleneck" observed in low-resource language processing.
- **mDeBERTa-v3-base** (`microsoft/mdeberta-v3-base`; He et al., 2023): Comprising 12 layers (13 hidden states), this model utilizes *Disentangled Attention*. Unlike the standard Transformer architecture which sums content and position embeddings, DeBERTa represents them separately. The attention weight $A_{i,j}$ between token i and j is calculated as:

3.2 Languages and Data

This study focuses on Spanish (*es*). We utilize a sentence level corpus derived from **Spanish Wikimedia**, containing 100,000 sentences to provide a broad representation of natural language patterns.

Gender Anchors: To define the semantic gender direction, we employ a set of definitional anchor pairs (e.g., *él/ella*, *hombre/mujer*). These pairs provide the reference points for calculating the gender subspace in the embedding layers (Bolukbasi et al., 2016; Zhao et al., 2019).

Profession Lists: To isolate morphosyntactic properties from latent societal bias, we organize two distinct target lists of profession terms:

- **Group A (Invariant):** Grammatically neutral terms that remain identical regardless of the referent's gender (e.g., *dentista*, *comerciante*, *estudiante*, *periodista*).
- **Group B (Marked):** Grammatically gendered terms that possess distinct masculine and feminine forms (e.g., *ingeniero/ingeniera*, *abogado/abogada*). These serve as a baseline to establish the expected magnitude of the gender signal when it is explicitly encoded in the surface form.

3.3 Context Sampling

For each target word w, we retrieve up to n sentences containing w via substring matching. We use $n = 20$ contexts per profession term and $n = 12$ contexts per anchor term across all languages, to ensure stable cen- troid estimation within CPU memory constraints. Words with fewer than the minimum available contexts are excluded.

3.4 Contextual to Static Reduction

Transformer models produce contextualised token representations that vary across inputs. To enable geometric analysis, we adopt the reduction framework of Bommasani et al., 2020. For each occurrence of word w in context c_i, subword representations are mean-pooled. The result is averaged across n contexts. Formally, at layer ℓ:

$$s_\ell(w) = \frac{1}{n} \sum_{i=1}^{n} h_\ell(w, c_i), \tag{1}$$

where $h_\ell(w, c_i)$ is the mean pooled subword representation. This yields one stable static embedding per layer for each target word.

3.5 Gender Direction Construction

Following Bolukbasi et al., 2016, we define a gender direction using predefined anchor sets. Let M and F denote the male and female anchor term sets. For each layer ℓ, anchor representations are reduced following Eq. (1), and the centroids for the male and female anchor sets (A_M and A_F) are defined as:

$$c_{M,\ell} = \frac{1}{|M|} \sum_{w \in M} s_\ell(w), \quad c_{F,\ell} = \frac{1}{|F|} \sum_{w \in F} s_\ell(w), \tag{2}$$

and the layer specific gender direction is:

$$g_\ell = c_{M,\ell} - c_{F,\ell}. \tag{3}$$

This vector encodes the principal geometric contrast between male and female anchors at layer ℓ (Bolukbasi et al., 2016).

3.6 Bias Measurement and Agreement

For each profession p and layer ℓ we compute two geometric estimators:

(a) **Gender Direction Projection (`proj`)**: The scalar projection of $s_\ell(p)$ onto the gender direction:

$$ProjBias_\ell(p) = \frac{s_\ell(p) \cdot g_\ell}{\|g_\ell\|}. \tag{4}$$

Positive values indicate alignment toward the male centroid; negative values indicate alignment toward the female centroid. The scalar projection preserves vector magnitude and is sensitive to the strength of the gender association (Bolukbasi et al., 2016).

(b) **Centroid Cosine Difference (`cosdiff`)**. As a complementary magnitude normalised estimator:

$$\text{CosBias}_\ell(p) = \cos(\mathbf{s}_\ell(p), \mathbf{c}_{M,\ell}) - \cos(\mathbf{s}_\ell(p), \mathbf{c}_{F,\ell}), \tag{5}$$

where cosine similarity is defined as (Manning et al., 2008):

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|}. \tag{6}$$

This estimator directly compares relative angular proximity to male and female centroids and is invariant to vector scale.

3.7 Estimator Agreement

To assess measurement robustness, we compute Spearman rank correlation (Spearman, 1904) between the two estimators across profession terms at each layer:

$$\rho_\ell = \text{Spearman}(\{ProjBias_\ell(p)\}_p, \{CosBias_\ell(p)\}_p). \tag{7}$$

$\rho\ell = 1$ indicates perfect ordinal agreement across professions.

3.8 Layer Wise Aggregation

For visualisation we compute the mean projection score across retained profession terms at each layer ℓ, yielding a layer wise bias trajectory:

$$\overline{\text{Proj}}_\ell = \frac{1}{|P|} \sum_{p \in P} \text{ProjBias}_\ell(p), \tag{8}$$

where P is the set of retained professions. This trajectory summarises how the overall geometric gender signal evolves across model depth.

4 Results

4.1 Estimator Agreement (RQ3)

Before interpreting bias scores, we verify measurement robustness. Table 1 reports the Spearman rank correlation ρ between the projection and cosine-difference estimators for each model across all layers. Every model achieves a minimum $\rho \geq 0.97$ and a mean above 0.99, showing a near perfect agreement. The one notable exception is XLM-V at layer 12, where ρ drops to 0.70, consistent with the representational collapse commonly observed at the final layer.

Table 1: Estimator agreement: Spearman ρ between projection and cosine-difference scores across layers (Spanish).

Model	**Min ρ**	**Max ρ**	**Mean ρ**
XLM-R Base	0.991	1.000	0.997
XLM-R Large	0.986	1.000	0.997
XLM-V Base	0.700	1.000	0.976
mDeBERTa-v3	0.967	0.999	0.990
mBERT	0.979	0.998	0.990

4.2 Group A: Invariant Professions (RQ1)

Table 2 reports projection scores for the ten invariant professions. We report measurements at each model's peak layer (e.g., Layer 9 for XLM-R Base) where bias magnitude is highest. As shown in Figure 3, computing a mean across all layers dilutes the signal because early layers focus on syntax and final layers prepare logits for output. The full layer by layer trajectories are provided in subsequent heatmaps.

Somey terms show neutral morphology: *dentista* (+0.01 in XLM-R Base) and *cantante* (−0.21 in XLM-R Large) remain near zero. However, other invariant terms exhibit strong male bias. For instance, *comerciante* scores +2.69 in XLM-R Base, while mDeBERTa classifies it neutrally (−0.03). *Piloto* shows consistently high male bias across all evaluated models.

4.3 Group B: Gendered Profession Pairs (RQ1)

Table 3 presents peak layer projection scores for the masculine and feminine forms of some of the gendered profession, alongside their magnitude asymmetry ($\Delta = |proj_M| - |proj_F|$). Figure 1 visualises this geometry for XLM-R Base.

While masculine forms align positively and feminine forms align negatively, the absolute magnitude of masculine scores is consistently higher. For example, in XLM-R Base at layer 9, *profesor* scores +3.45 while *profesora* scores +0.08 ($\Delta = 3.37$). This asymmetry indicates that masculine forms serve as the

Table 2: Projection scores for Group A (invariant professions) at each model's peak layer. Positive values indicate male bias.

Profession	XLM-R B (L9)	XLM-R L (L16)	XLM-V (L7)	mDeBERTa (L5)	mBERT (L11)
policía	+2.09	+0.85	−0.77	−0.22	+1.12
artista	+3.08	+2.14	+1.49	+0.64	+5.04
comerciante	+2.69	+1.41	+2.36	−0.05	+3.71
periodista	+2.56	+1.03	+0.81	+0.26	+3.13
piloto	+3.33	+1.78	+2.03	+1.22	+4.72
cantante	+2.15	−0.21	+0.01	+0.31	+0.36
juez	+2.56	+1.96	+1.19	+0.76	+3.32
dentista	+2.08	+1.41	+1.02	+0.19	+3.00
estudiante	+2.40	+1.74	+1.35	+0.76	+4.43

default geometric anchor for the profession concept, effectively treating feminine forms as semantic deviations. Figure 2 further demonstrates that this massive disparity between masculine and feminine projections is not an isolated artifact of the peak layer, but a persistent mathematical reality across the entire depth of the architecture.

Table 3: Peak layer projection scores for Group B gendered pairs (XLM-R Base, Layer 9). $\Delta = |proj_M| - |proj_F|$. Positive Δ indicates masculine dominance.

Profession	Masculine	Feminine	Δ
abogado/a	+3.48	+0.68	+2.80
profesor/a	+3.45	+0.08	+3.37
mecánico/a	+2.94	+1.15	+1.80
cocinero/a	+2.90	+0.56	+2.34
minero/a	+2.66	+0.90	+1.76
arquitecto/a	+3.06	+1.52	+1.54
ingeniero/a	+3.15	+0.88	+2.27

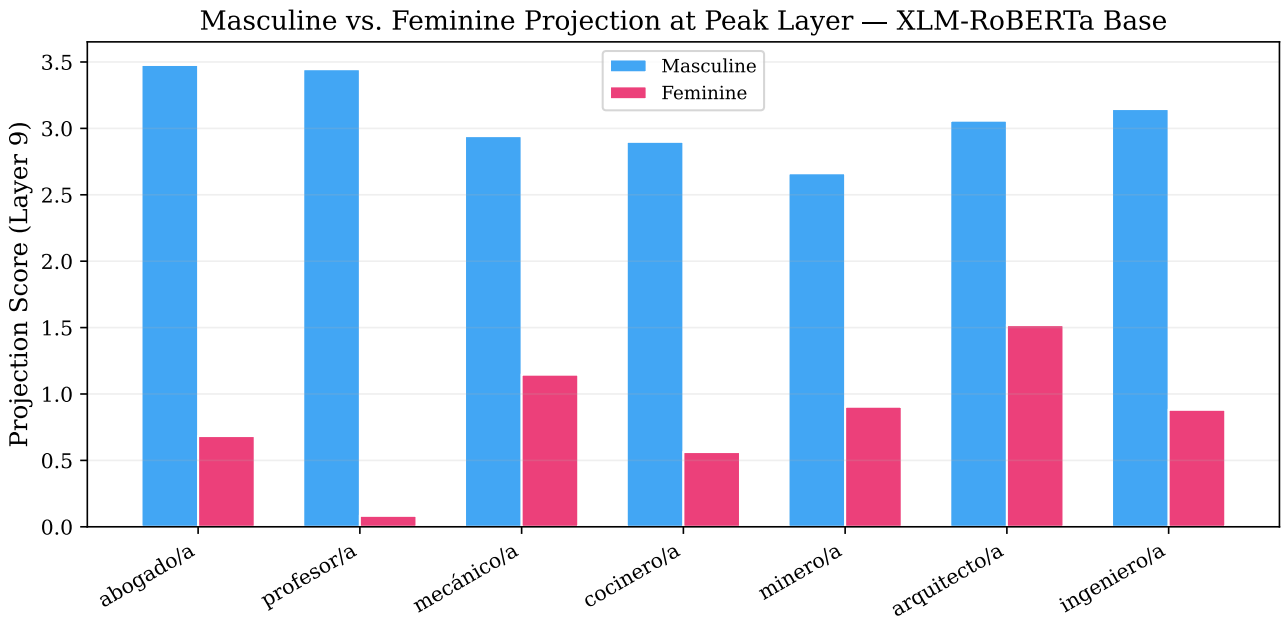

Figure 1: Masculine vs. feminine projection scores at peak layer 9 for XLM-RoBERTa Base.

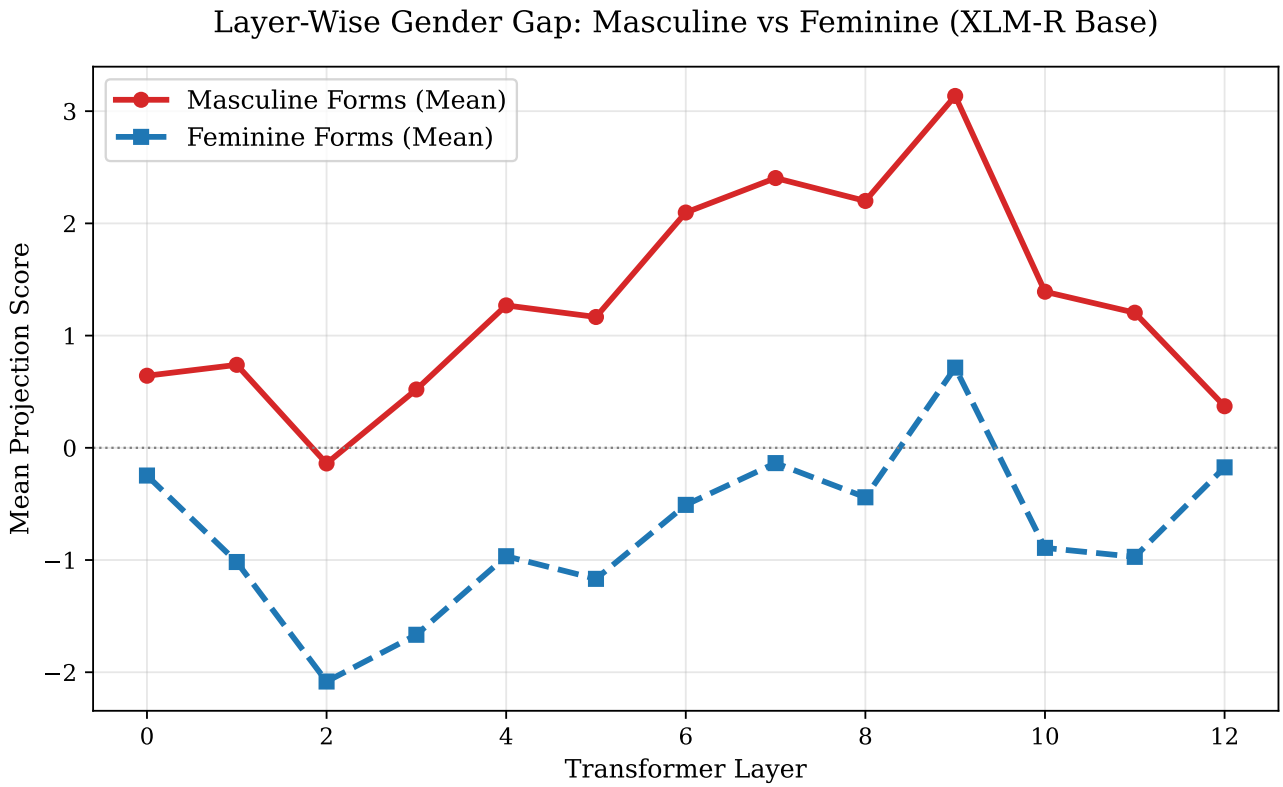

Figure 2: Layer wise mean projections for masculine vs. feminine forms in XLM-RoBERTa Base. The masculine forms persistently dominate the semantic space across all network depths.

4.4 Layer Wise Bias Trajectory (RQ2)

Figure 3 displays the mean projection score across representations at each transformer layer. XLM-R Base, XLM-R Large, and XLM-V share a rise peak decay trajectory: bias is weak initially, peaks in the middle layers (7 to 16), and decreases toward output generation. This matches observations that middle layers encode core semantic relationships.

In contrast, mDeBERTa-v3-base never develops the male biased signal seen in RoBERTa models but instead, its projection remains near zero or slightly negative across all layers. This architectural shift is analysed in Section 5.

4.5 Architecture-Driven Direction Reversal (RQ4)

Figure 4 contrasts per profession heatmaps for XLM-R Base and mDeBERTa-v3. While XLM-R Base displays positive male values in middle layers, mDeBERTa projects predominantly negative or near zero values.

Since both models use comparable training corpora, this results in a reversal likelihood from mDeBERTa's *Disentangled Attention* mechanism. By separating content and positional matrices, the model mitigates the accumulation of gender associations from grammatical context into the core representations.

The full per-profession heatmap for XLM-R Base (Figure 5) illustrates the layered structure: Group A terms cluster near the boundary between red and blue, while Group B masculine terms are deep red and feminine terms are deep blue.

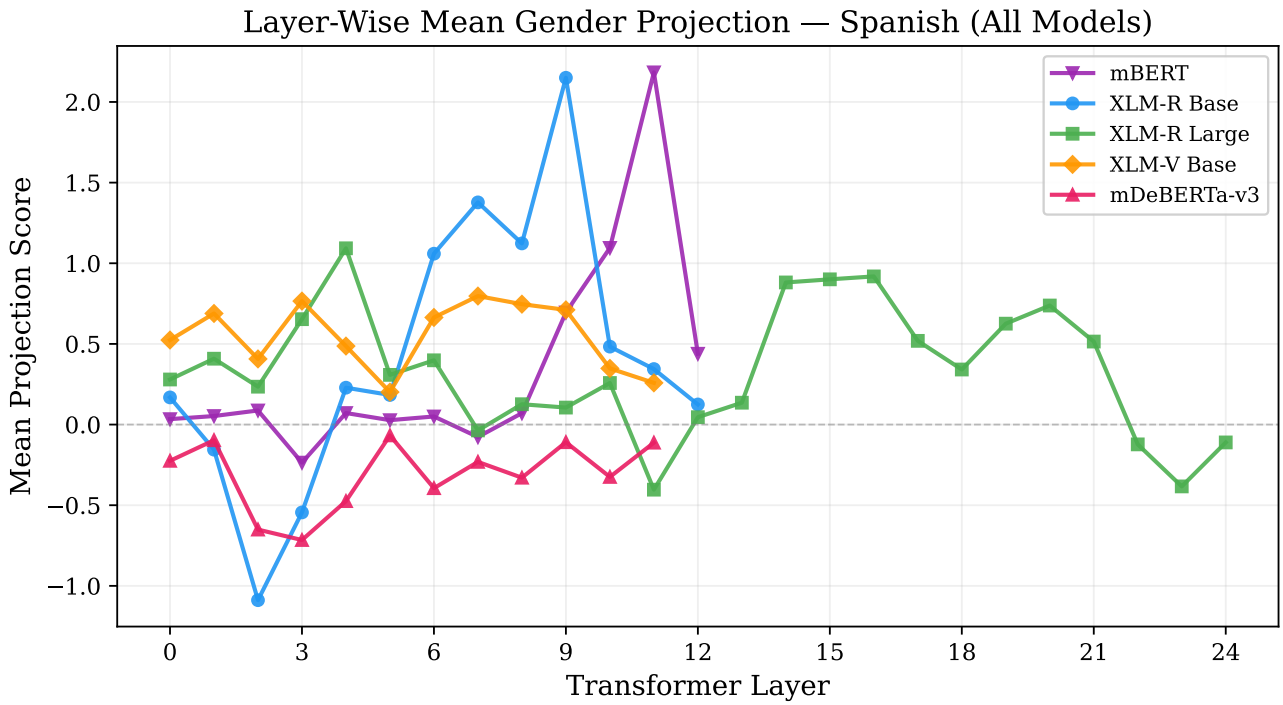

Figure 3: Layer wise mean gender projection across all professions for four Spanish models. The three RoBERTa family models show a rise, peak and decay pattern while mDeBERTa-v3 stays near or below zero.

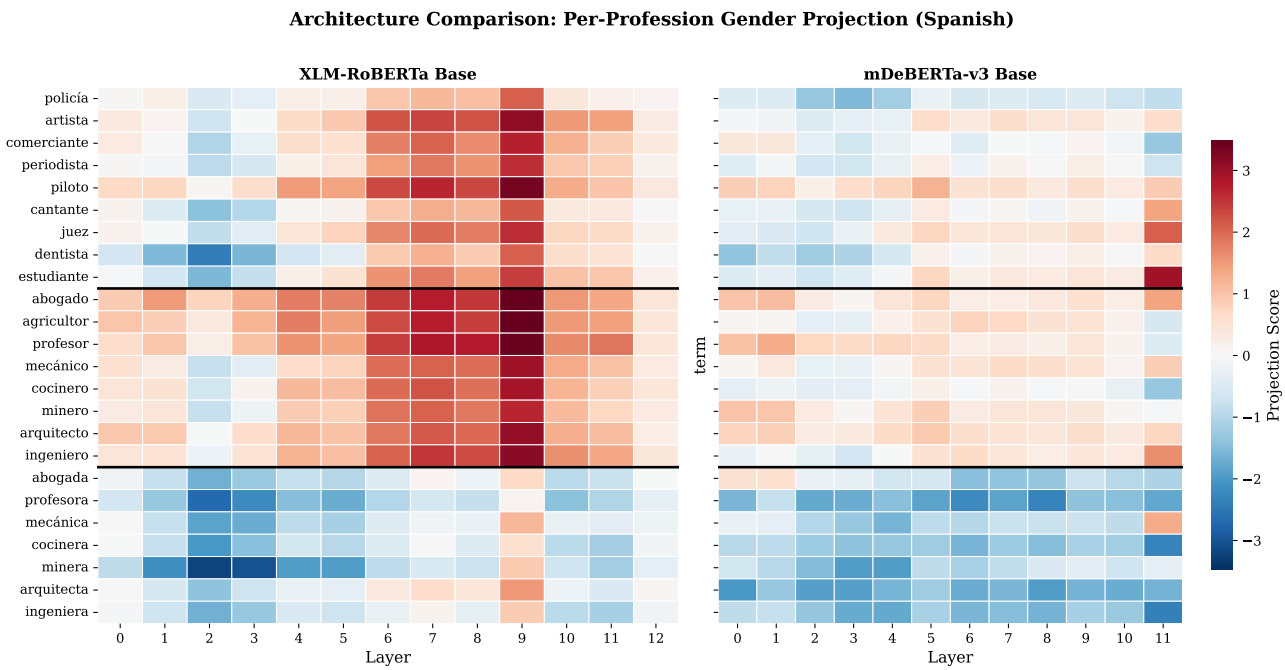

Figure 4: Per profession projection heatmaps: XLM-RoBERTa Base (left) vs. mDeBERTa-v3 (right). Disentangled Attention produces a reversal in bias direction across all professions. Layer 12 is excluded for mDeBERTa due to ELECTRA-head artifacts.

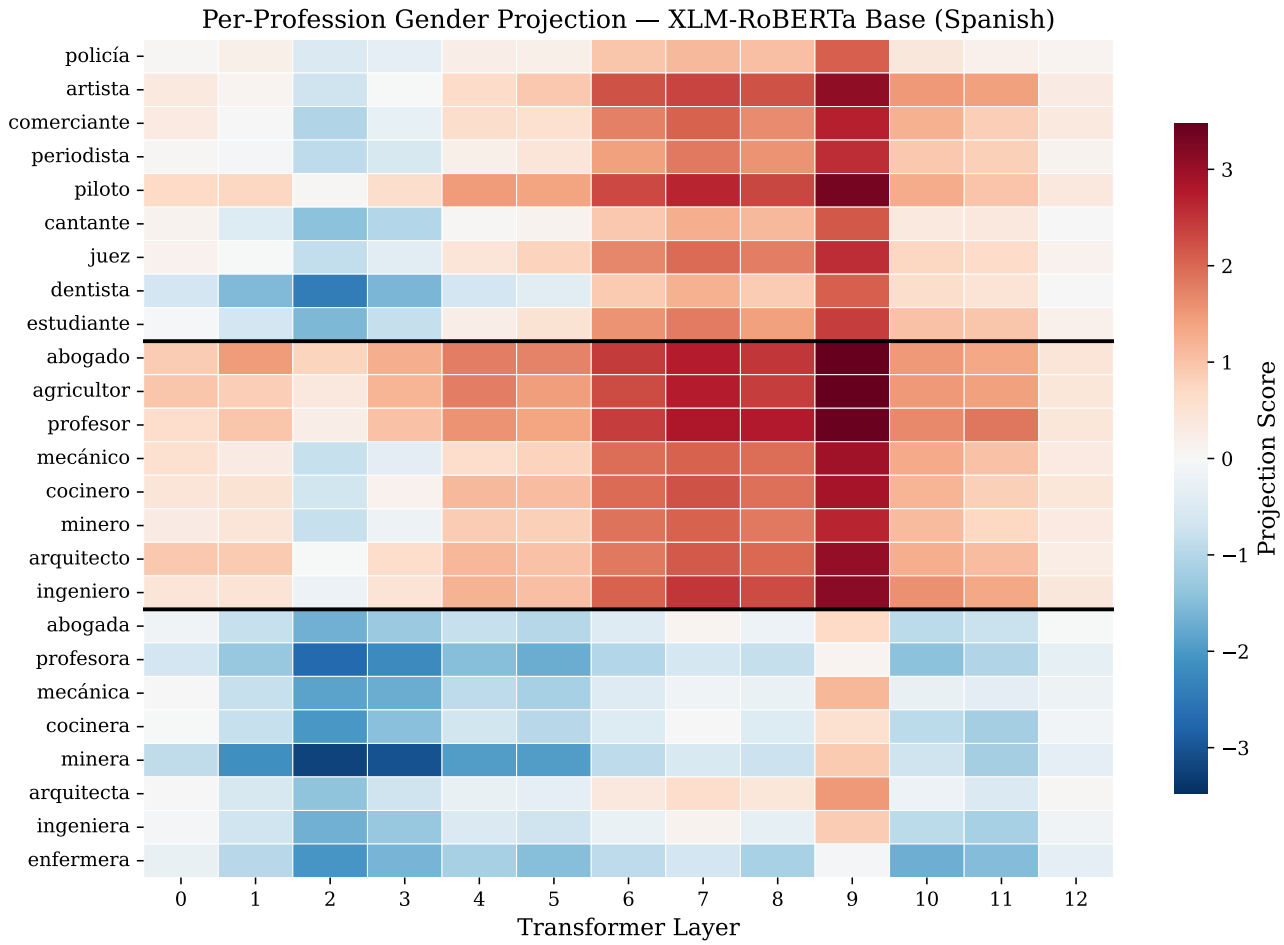

Figure 5: Full per profession heatmap for XLM-RoBERTa Base. Horizontal lines separate Group A (invariant), Group B masculine, and Group B feminine professions.

4.6 Cross Lingual Patterns (RQ1)

To contextualise these results, Figure 6 maps the layer wise projection for XLM-RoBERTa Base across Spanish, Arabic, and Tigrinya, enabling a controlled comparison. Table 4 provides key metrics.

Despite large disparities in pretraining corpus size and morphological structure, the bias trajectory is highly consistent. For all three languages, XLM-R Base produces a smooth rise that peaks precisely at Layer 9, with magnitudes bounded between +2.15 and +2.99. This indicates that the base architecture structures gender bias uniformly across resource levels.

Table 4: Cross lingual summary: layer wise projection statistics for 12-layer models.

Lang.	**Model**	**Peak Layer**	**Peak Proj.**	**Min** ρ	n (prof.)
Spanish	XLM-R Base	9	+2.15	0.991	25
Arabic	XLM-R Base	9	+2.64	0.964	23
Tigrinya	XLM-R Base	9	+2.99	0.992	30

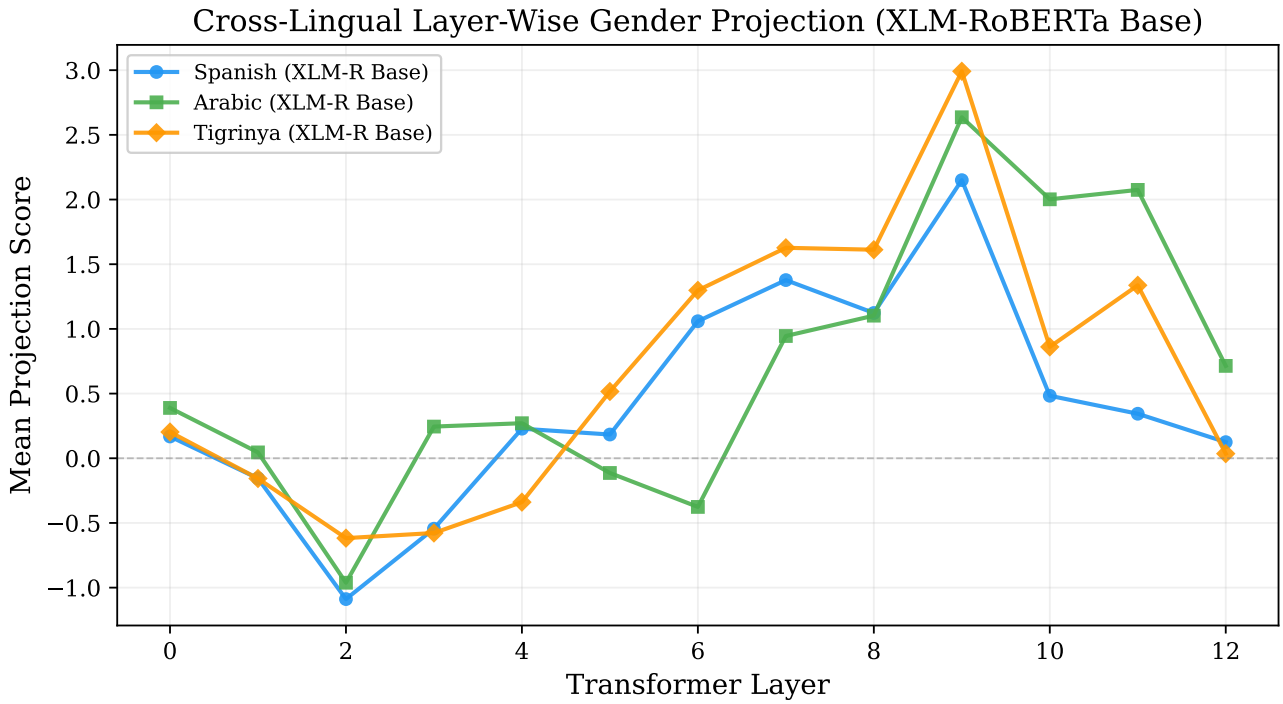

Figure 6: Cross lingual layer wise mean projection for XLM-RoBERTa Base across Spanish, Arabic, and Tigrinya. Remarkably, the architecture encodes bias with an identical structural trajectory (peaking exactly at Layer 9) regardless of the language's resource level.

5 Discussion and Conclusions

5.1 Grammatical Neutrality vs. Societal Frequency

Multilingual transformers learn morphological neutrality without explicit supervision, locating invariant professions (*dentista*, *cantante*) near the geometric origin. However, statistical frequency in the pretraining corpus overrides grammatical rules for historically male dominated roles. For example, XLM-R Base projects the invariant *comerciante* (+2.69) similarly to the explicitly masculine *mecánico*

(+2.94). Only mDeBERTa-v3 suppresses this effect (−0.03), indicating that architectural design also has an influence in the conflict between grammar and frequency.

5.2 The Masculine Default

The asymmetry analysis shows that masculine forms occupy a dominant position as the default representation of a profession. Feminine forms correctly orient toward the female centroid but remain closer to the origin. By preserving this data asymmetry or imbalance, the models treat masculine professions as more semantically central, carrying practical implications for certain applications like search retrieval or automated text analysis, and the interpretability research.

5.3 Architecture over Pretraining Data

Our results emphasize architecture as a dominant factor in bias encoding. mDeBERTa-v3 reverses the bias direction relative to RoBERTa-family models, preserving relative profession rankings ($\rho > 0.97$) but flipping their geometric orientation. We attribute this to Disentangled Attention, by separating content from positional context, the model prevents societal frequencies from having too much influence on the gender direction.

Similarly, the cross lingual analysis shows that semantic structures depend less on pretraining volume than often assumed. XLM-R Base encodes bias with an identical structural trajectory (peaking at Layer 9) across Spanish, Arabic, and Tigrinya, despite massive disparities in corpus size. This suggests the attention mechanism functions as a universal template for how abstractions like gender are structured.

5.4 Final Remarks and Limitations

Since we only looked at base embeddings without fine tuning, it's possible these bias patterns could change once the models are adapted for specific tasks. Moreover, our anchor sets and Spanish context corpus may not capture all domains of language use.

In conclusion, multilingual transformers encode measurable geometric gender bias that peaks in the middle semantic layers. Evaluating fairness exclusively at the final layer might underestimate magnitude. Our results suggest that measuring geometric structures across layers is more reliable. Furthermore, architectural choices like attention factorisation seem to inherently mitigate social bias, bypassing the need for explicit debiasing pipelines.

Reproducibility Statement

To facilitate full reproducibility, the source code, data parsing scripts, and complete evaluation pipeline used in this study are publicly available in our GitHub repository: https://github.com/JuanArango99/ml4nlu-gender-bias.

Appendix A. Notable Individual Model Trajectories

While the main text aggregates layer wise projections or restricts analysis to XLM-RoBERTa Base to uncover structural consistency, individual models exhibit distinct edge case behaviors. We highlight four examples below.

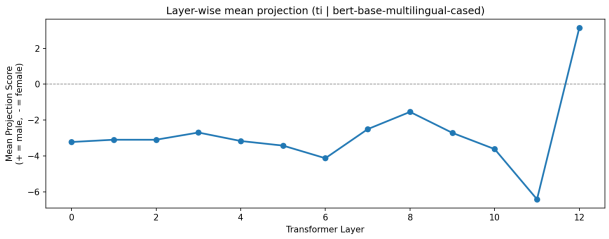

Figure 7: Tigrinya (mBERT): Extreme low-resource collapse. With insufficient pretraining data, mBERT fails to generate a societal frequency signal, collapsing into arbitrary negative projections.

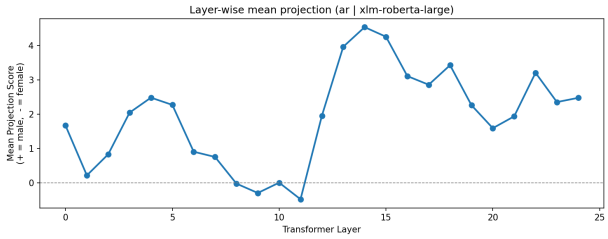

Figure 8: Arabic (XLM-R Large): Late stage peaking. While Base models peak near layer 9, this Large model peaks at layer 14 with a +4.54 magnitude.

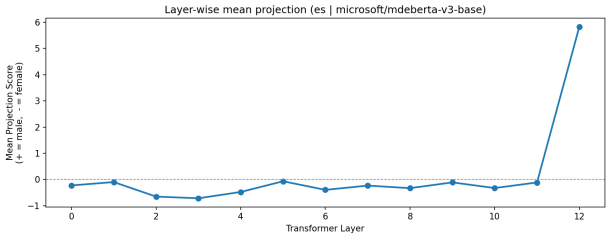

Figure 9: Spanish (mDeBERTa-v3): Disentangled Attention reversal. The projection score stays negative across all middle layers, reversing the RoBERTa upward curve.

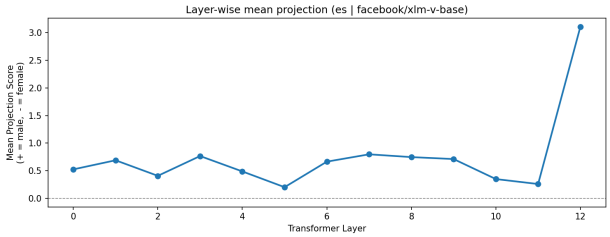

Figure 10: Spanish (XLM-V Base): Final layer artifact spike. The bias score exhibits an unnatural spike at layer 12 (+3.11) possibly due to token level output preparation.

Appendix B. Evaluated Lexicons (Spanish)

This section details the explicit Spanish word sets used to construct the gender subspace and evaluate profession embeddings. The anchor sets (Table 5) capture fundamental gender semantics, while the profession sets (Table 6) are divided into morphologically invariant (Group A) and gender-inflected (Group B) terms.

Table 5: Spanish gender anchor sets used to define the semantic subspace.

Masculine Anchors	**Feminine Anchors**
hombre	mujer
niño	niña
padre	madre
él	ella
hermano	hermana
esposo	esposa
abuelo	abuela
hijo	hija
tío	tía
varón	señora
chico	chica

Table 6: Spanish profession terms evaluated in this study, categorized by morphological group.

Group A (Invariant)	**Group B (Masculine)**	**Group B (Feminine)**
policía	abogado	abogada
recepcionista	enfermero	enfermera
artista	agricultor	agricultora
comerciante	profesor	profesora
periodista	mecánico	mecánica
piloto	cocinero	cocinera
cantante	limpiador	limpiadora
juez	minero	minera
dentista	arquitecto	arquitecta
estudiante	ingeniero	ingeniera

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